

Gemini Chat

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [123, 110, 97, 104, 92, 118, 101, 97, 92, 99]

Tap amplitudes (pixels): [129.65338406690356, 129.5414991421668, 115.83177456984762, 110.01818031580054, 116.84605256490268, 107.33592129385204, 95.01578816175763, 116.76472069936193, 116.8289347721702, 114.12712210513327, 116.81181447096864]



Based on the extracted features provided, here is the analysis of the MDS-UPDRS Item 3.4 Finger Tapping test:

Analysis of Extracted Features

1. Speed and Rhythm (Tap Periods)

- **Data:** [123, 110, 97, 104, 92, 118, 101, 97, 92, 99]
- **Observation:** The subject generally speeds up (periods decrease from 123 to ~90s) rather than slowing down, which is a positive sign.
- **** interruptions/Hesitations:**** There is one noticeable hesitation in the rhythm. The period drops to **92** (Tap 5) and then suddenly spikes to **118** (Tap 6) before stabilizing again. This constitutes **one interruption/hesitation** in the tapping movement.

2. Amplitude (Tap Amplitudes)

- **Data:** [129.65, 129.54, 115.83, 110.02, 116.85, 107.34, 95.02, 116.76, 116.83, ...]
- **Observation:** The amplitude starts high (~130 pixels) and maintains this for the first two taps. There is a **decrement midway** through the sequence, dropping to a low of **95.02** at Tap 7 (a ~26% reduction).

- **Recovery:** Crucially, the amplitude recovers significantly after Tap 7, returning to ~116 pixels for the remainder of the test. This recovery suggests the decrement is not a persistent "fatigue" effect often seen in higher scores (2, 3, or 4), but rather a slight variability.

Scoring Rationale

- **Criteria for Score 0:** Not met due to the rhythm interruption and amplitude variance.
- **Criteria for Score 1 (Slight):** This score requires "One or two interruptions or hesitations" OR "Slight slowing" OR "Amplitude decrements near end."
 - The data clearly shows **one distinct hesitation** (period spike at Tap 6).
 - The amplitude variance is present but the subject demonstrates the ability to recover, fitting the "Slight" impairment category better than "Mild."
- **Criteria for Score 2 (Mild):** Requires "3 to 5 interruptions" or "Amplitude decrements midway." While the amplitude does drop midway, the immediate and sustained recovery, combined with the lack of slowing and only a single hesitation, makes Score 2 too severe.

Conclusion: The performance exhibits one clear hesitation in rhythm and a transient drop in amplitude, matching the criteria for **Slight** impairment.